

DAVID ALEXANDER HILL

AI Automation & Development

CONTACT

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Johannesburg
South African Citizen

EDUCATION

Matric
Cedarwood College, 2016

TECHNICAL SKILLS

Multi-Agent AI Systems (OpenClaw)
Agentic System Architecture
MCP (Model Context Protocol)
MCP Server Development
LLM Integration & Prompt Engineering
RAG & Vector Databases
ChromaDB / Vector Memory Systems
AI Application Development
Anthropic API / Claude
Python, JavaScript, TypeScript & PowerShell
Node.js & Playwright Automation
HTML, CSS & API Integration

LANGUAGES

English — Native

CERTIFICATIONS

Kaspersky Sales Specialist · Barracuda
Certified Sales Partner

INTERESTS

Skateboarding, PC hardware, gaming and video editing

PROFILE

AI engineer with hands-on experience building multi-agent AI systems, LLM-based applications and retrieval-backed memory architectures. Designs system architecture and agent behaviour, and owns the design decisions, code review, debugging and production validation. Self-hosts a personal multi-agent AI stack in daily production use, and led the deployment of a production multi-agent system for a client workforce. Coding with AI tools since 2022, from early manual prompt loops through to modern agentic development.

EXPERIENCE

Neozeit Nov 2023 – Present
IT Engineer — Development & Automation

- Led the design and deployment of a multi-agent AI automation system (OpenClaw) for a two-company client workforce, including integration of MemPalace, a ChromaDB-based memory system for persistent agent context.
- Designed a tiered model-delegation strategy for cost control across the agent fleet: reasoning models for planning and judgment, lightweight models for mechanical execution, and direct scripts for bulk data work.
- Built a custom ClickUp MCP server (TypeScript, 18 tools) for live task and project status, integrated into the flagship production agent.
- Built and deployed production web applications for clients, including a doctor location mapping tool (Node.js).
- Developed and maintain a financial market dashboard for a financial advisory practice with automated Telegram delivery of daily updates, built for Neozeit client work using a custom Telegram bot.
- Built a Microsoft 365 recon and visibility dashboard using Extractor Suite and Analyzer Suite outputs for SharePoint environments and tenant-wide security statistics.
- Infrastructure and support administration across Neozeit's SME client portfolio, including Microsoft 365, Active Directory, and Proxmox VE cluster management.
- Sole operator of Neozeit's Proxmox VE infrastructure (3-node cluster, 12-drive RAID6 backup server, LTO-4 tape rotation, managed Avaya switch).
- Built and configured a Windows Server 2025 VM on Neozeit's Proxmox infrastructure for a medical client, including networking and user/group setup, then migrated all of the client's company data sets onto it and validated end-to-end that users could log in and work; an external vendor handled the Pastel application install and licensing.
- Built shell-script automation for update and patch management across a client's macOS fleet of roughly 160 devices.
- Promoted from Junior to IT Engineer after 4 months, recognised for operating consistently above entry level across all project areas.

Drago Technologies Oct 2022 – Feb 2023
IT Technician

- Delivered first-line IT support for business clients, resolving hardware, software and connectivity issues.
- Assisted with desktop and laptop configuration, OS deployment and peripheral setup across client environments.

TGVision Oct 2021 – Aug 2022
TV Technician

- Diagnosed and repaired flat-panel televisions across a range of brands and screen technologies.
- Performed backlight array replacement, CoF (Chip on Film) bonding, screen replacement, capacitor replacement and mainboard repair.
- Managed repair workflow from intake through diagnosis, parts sourcing, repair execution and quality assurance.

AVeS Cyber Security Jun 2019 – Aug 2019
Intern — Technical Administrator

- Provided IT hardware and software troubleshooting support internally and for clients.
- Assisted account managers by contacting clients and communicating product and package benefits.

- Gained foundational exposure to cybersecurity tooling and the vendor ecosystem, directly informing later Kaspersky and Barracuda certifications.

Earlier Roles 2016 – 2018

Au Pair · Paintball Marshal · Retail Cashier (Part-Time)

- Private Family (2018): daily transport and homework support for a school-age child. Battlezone Paintball (2017-18): safety briefings, game flow and equipment maintenance. Riverside Liquors (2016-17): PoS operation, sales and stock queries.

KEY PROJECTS

Personal Multi-Agent AI Stack

Architected and self-host a production multi-agent system on OpenClaw for daily personal and family use: three isolated agents (a primary personal assistant, a sports-science agent used live by a professional cyclist's own training business, and a family-support agent), all on Kimi K3 with a GLM-5.2 fallback via OpenRouter. Scheduled background jobs follow the same tiered-delegation principle as the client deployment: lightweight recurring tasks run locally on gemma4:12b, jobs needing real judgement escalate to GLM-5.2. Each agent runs its own isolated MemPalace — an open-source ChromaDB-based memory system exposing 30 tool functions via its MCP server — for persistent, agent-scoped memory with zero cross-contamination. Self-hosted with no external dependencies for core execution.

Multi-Tenant Agent Deployment

Designed and deployed a production multi-agent system on OpenClaw for a client workforce across two tenant companies: an orchestrator, a live production executive-assistant agent, and two department agents (marketing and social media) designed and scoped for rollout. Integrated MemPalace, a ChromaDB-based memory system with physically separate palace instances and wing-scoped isolation per tenant. Tenant and department boundaries are enforced by filesystem and tool-schema configuration, not just documented policy, backed by a gated knowledge-graph write protocol and automated diary, compression and promotion pipelines. Migrated the agent fleet off the Anthropic API to Kimi K3 in July 2026, cutting inference costs while enhancing reliability, keeping the architecture provider-agnostic by design. Ongoing development and maintenance.

AI Document-Intelligence Agent (Upcoming)

Confirmed upcoming engagement for a client, not yet started: an agent that reads Excel spreadsheets, invoices and PDFs and pre-populates Xero accounting entries, reducing manual data capture in bookkeeping workflows.

Vantage Point Dashboard — AI Cyber Intelligence & GTM Platform (Cybersecurity Vendor)

Full-stack platform combining RSS threat-intel aggregation, autonomous competitor research and persona-tailored content generation. Custom ReAct-style agent loop built directly against the Anthropic Messages API — no agent framework — with end-to-end SSE streaming of agent reasoning and tool calls to a live UI. Deliberately tiered: deterministic keyword heuristics on the high-volume path, model calls reserved for curated work. React, TypeScript, Fastify, Prisma, PostgreSQL. Build complete and awaiting client testing and feedback.

Doctor Location Mapping Tool (Node.js)

Production mapping platform for a medical client. Directors and marketing teams use it to visualise which doctors work with the organisation and where they are located. Node.js, Express, Sequelize, MariaDB, Leaflet, with React web, React Native Android and a PWA build. Ships a read-only MCP server exposing codebase query tools.

Automated Inspection Pipeline (Medical Client)

Synthetic-monitoring pipeline (Python, Playwright) running scripted transactions against a medical client's production website and patient-booking platform on a systemd-timer schedule. Deterministic checks catch the common case at zero model cost. A failure triggers a cheap model (GLM-4.5-Air) for triage, and a full agent turn only runs when the failure needs a human-readable report, which then gets delivered automatically. Same tiered-delegation approach used across the rest of the agent work, applied here to uptime and performance monitoring instead of task orchestration.

MarketPulse Financial Market Dashboard

Financial market dashboard for a financial advisory practice, with automated Telegram delivery of daily updates, built for Neozeit client work using a custom Telegram bot.

SELF-DIRECTED DEVELOPMENT (2022 – PRESENT)

- Began coding with AI assistance in 2022, using early-generation LLMs in a manual prompt-and-paste workflow before agentic coding tools existed. Now works fluently across modern AI development tooling including Claude Code and Cursor.
- Self-studied Python, JavaScript, HTML and CSS through hands-on project work.
- Built service health monitoring scripts with automated alerting for personal web services.
- Developed data-fetching and categorisation scripts to organise and process structured datasets.
- Active GitHub portfolio at github.com/SassBlanket98 with live project repositories.